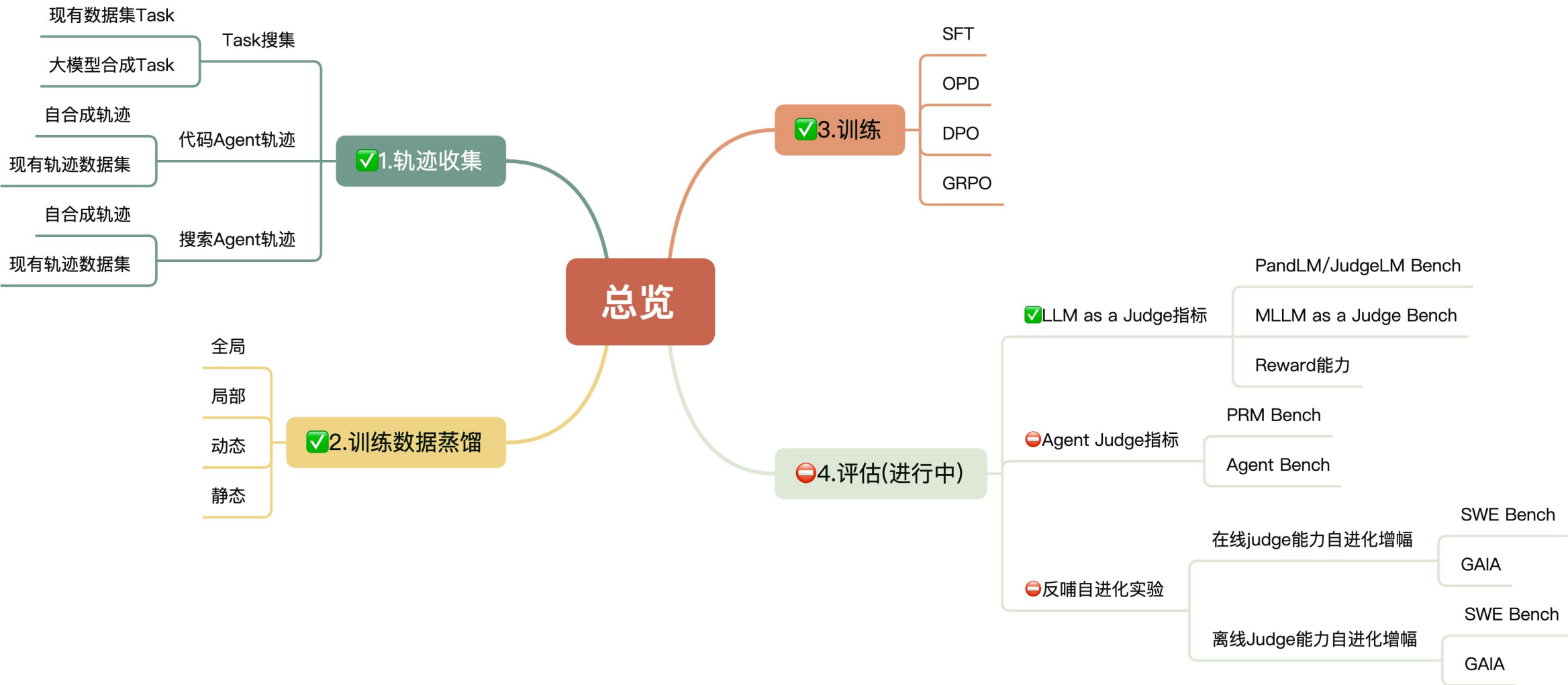


04

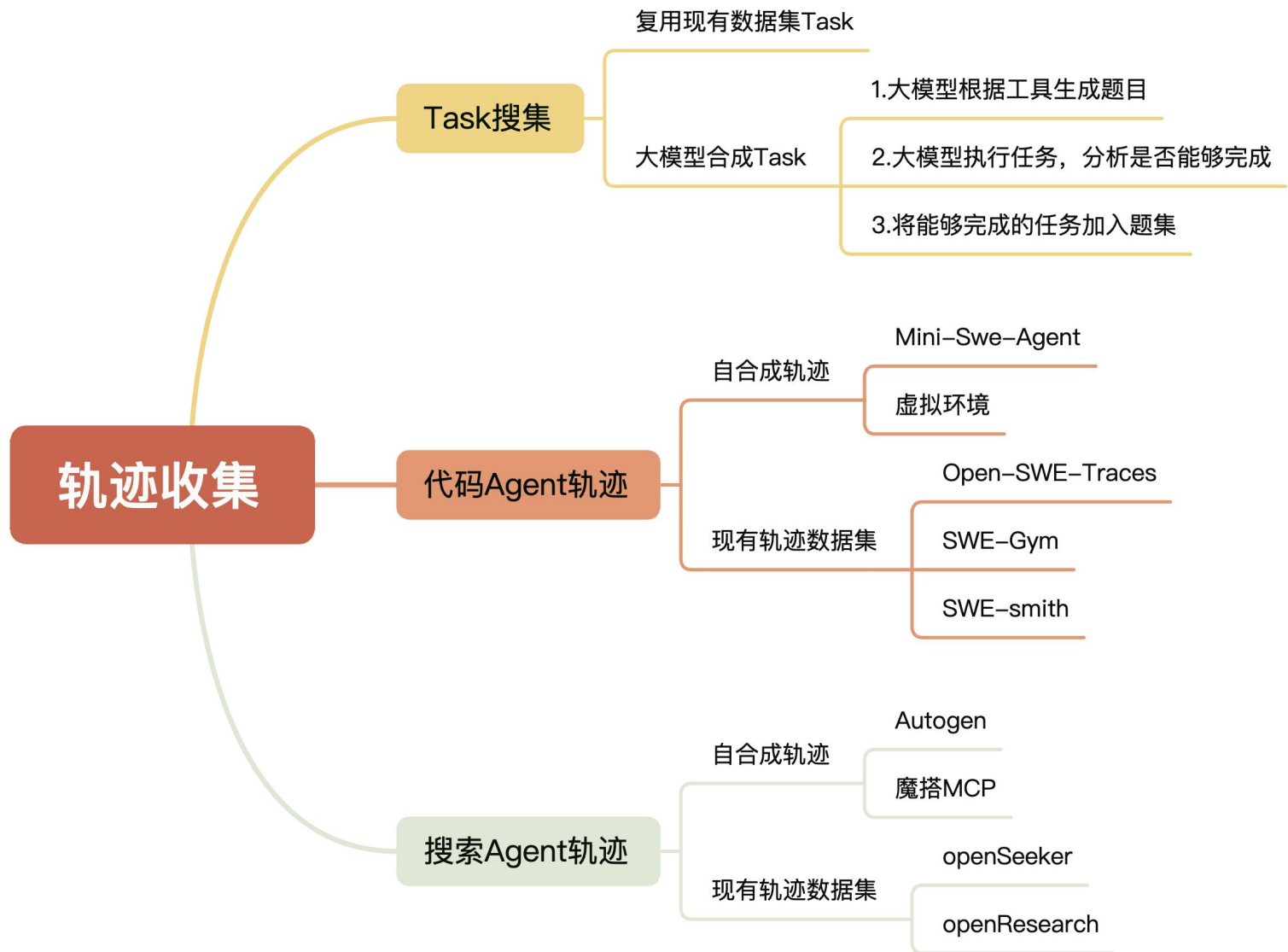
方案执行2



●4.1 执行总览



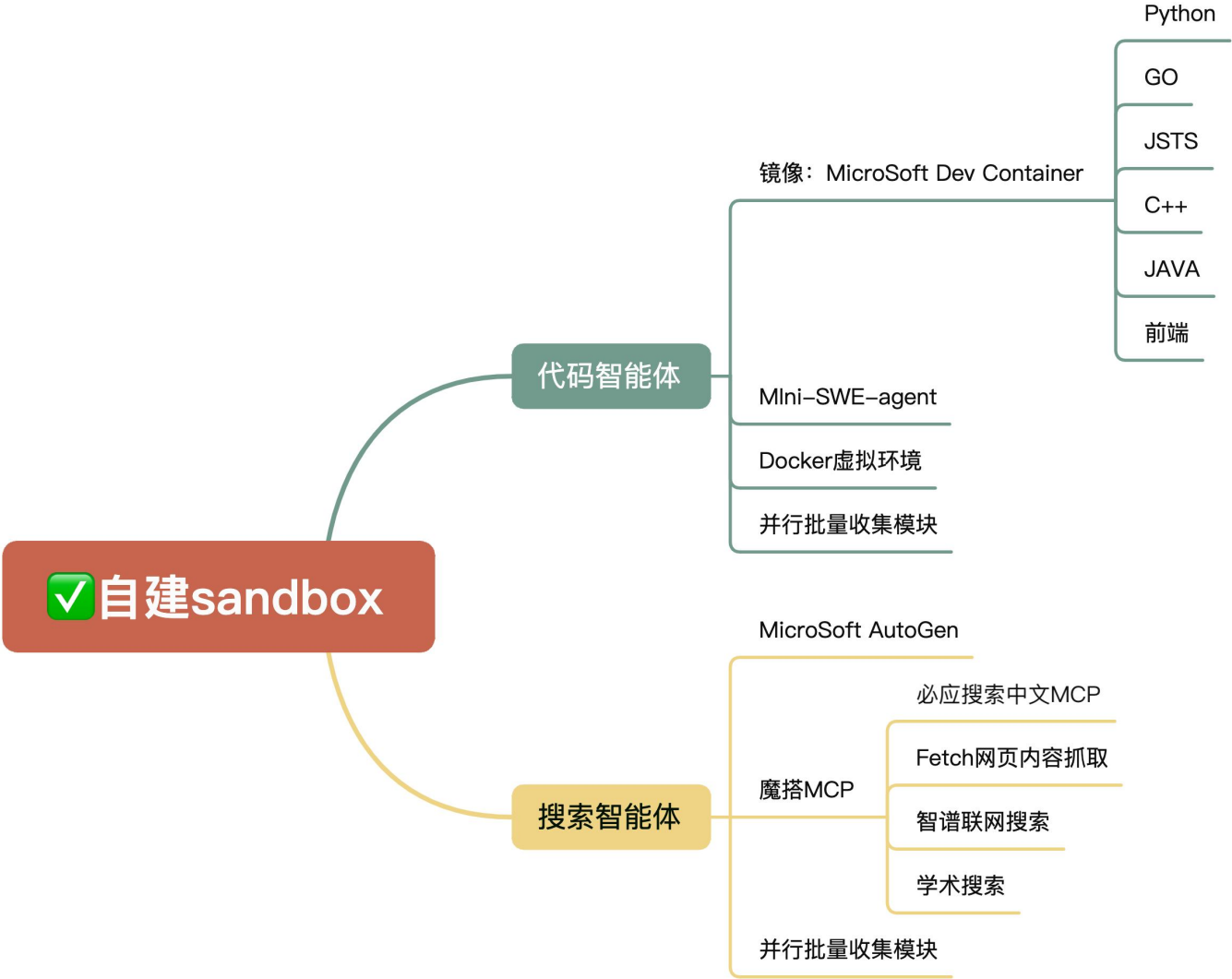
●4.2 轨迹收集



●4.2 SandBox搭建

维度	工作流 Workflow	Agent
核心定义	预先设计好的固定流程	能根据目标自主决策和调整行动
执行路径	通常是确定的、静态的	通常是动态的、可变化的
控制方式	人提前规定每一步怎么走	模型根据当前状态决定下一步
灵活性	较低，适合标准化任务	较高，适合开放式任务
任务拆解	通常由开发者手动拆好	Agent 可自行规划、拆解、反思
工具调用	工具调用顺序通常固定	工具调用由模型动态选择

●4.2 SandBox搭建



●4.2 SandBox搭建

```
O Trajectory 1/3 - swe-agent__test-repo-1.traj.json - Step 1/10

SYSTEM

You are a helpful assistant that can interact with a computer.

Your response must contain exactly ONE bash code block with ONE
command (or commands connected with && or ||).
Include a THOUGHT section before your command where you explain your
reasoning process.
Format your response as shown in <format_example>.

<format_example>
Your reasoning and analysis here. Explain why you want to perform the
action.

```bash
your_command_here
```

</format_example>

l Step++ h Step-- 0 Step=0 $ Step=-1 j Scroll down k Scroll up ^p palette
```

代码智能体轨迹

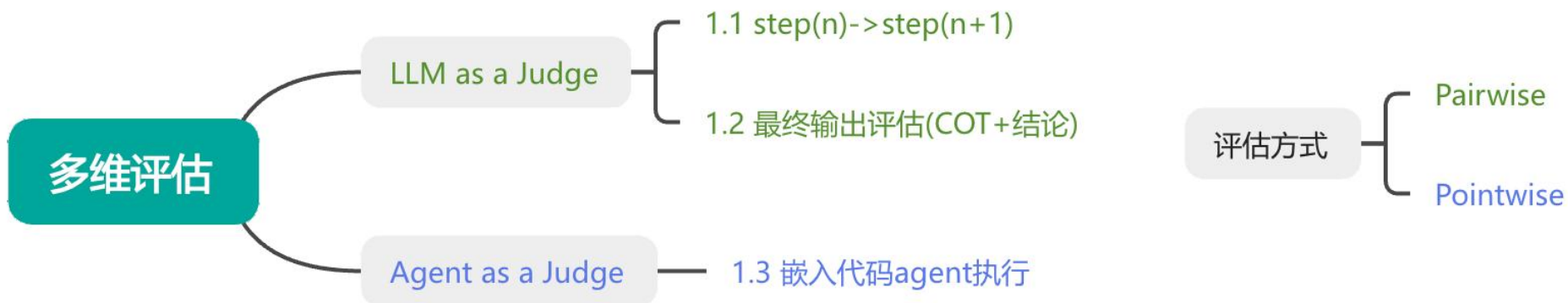
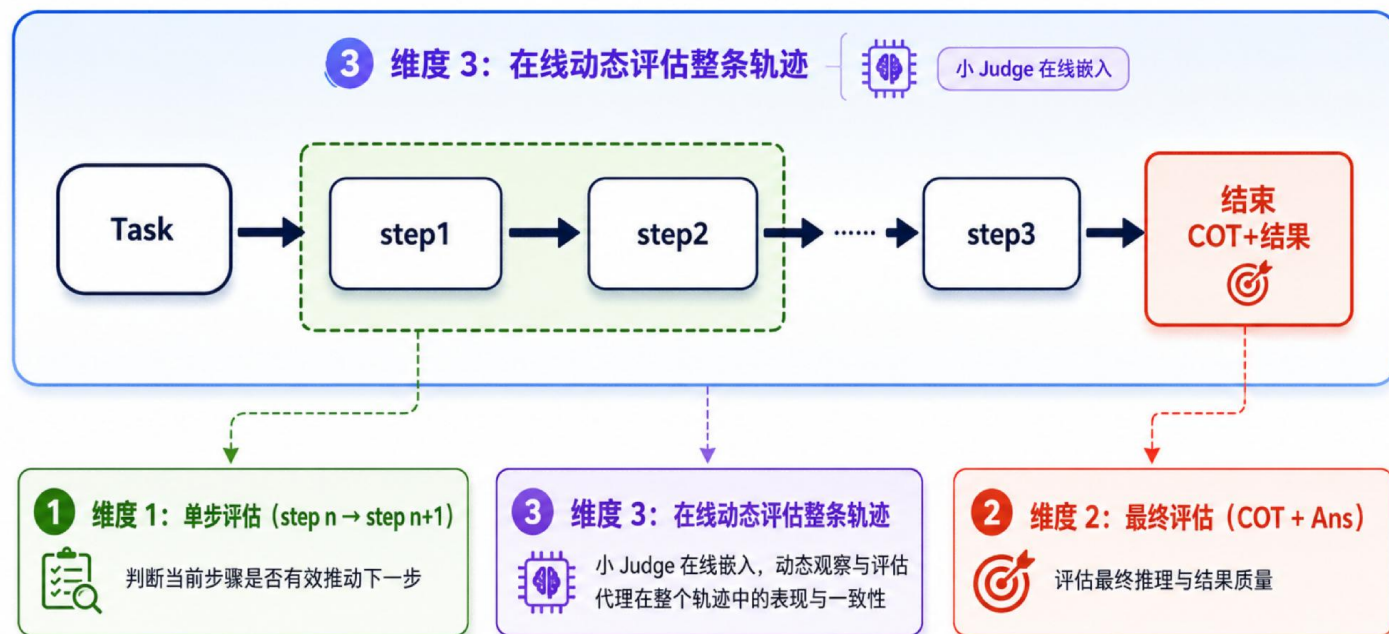
```
"models_usage": null,
"metadata": {},
"content": "}\n</tool_call>",
"type": "ThoughtEvent"
}
内容: }
</tool_call>
来源: fetch_agent

----- 消息 3 -----
类型: ToolCallRequestEvent
完整消息结构:
{
  "source": "fetch_agent",
  "models_usage": {
    "prompt_tokens": 478,
    "completion_tokens": 48
  },
  "metadata": {},
  "content": [
    {
      "id": "019f23bbe9bd58efe98ccdeb33e1ae7f",
      "arguments": " {\\"url\\": \\"https://www.douban.com\\", \\"max_length\\": 5000, \\"raw\\": false, \\"star"
      "name": "fetch"
    }
  ],
  "type": "ToolCallRequestEvent"
}
内容: [FunctionCall(id='019f23bbe9bd58efe98ccdeb33e1ae7f', arguments=' {\\"url\\": "https://www.douban.com"
来源: fetch_agent

----- 消息 4 -----
类型: ToolCallExecutionEvent
完整消息结构:
```

搜索智能体轨迹

●4.2 训练数据构造-SFT数据



●4.2 训练数据构造-DPO



●4.2 训练数据构造-GRPO



Pairwise Correctness Reward

$$R_{\text{pair}} = 1, \text{ if } \hat{y}_{\text{pair}} = y_{\text{pair}}^*; \\ -1, \text{ otherwise}$$

判断与正解一致则奖励



Pointwise Calibration Reward

$$R_{\text{point}} = 1 - 2 |s - s^*|$$

单点评分越接近标准分越好



Pointwise-Pairwise Consistency Reward

$$y_{\text{pw}} = A, \text{ if } s_A - s_B > \delta; \\ y_{\text{pw}} = B, \text{ if } s_B - s_A > \delta; \\ y_{\text{pw}} = \text{tie}, \text{ otherwise}$$

$$R_{\text{cons}} = 1[y_{\text{pw}} = \hat{y}_{\text{pair}}] \cdot 1[\hat{y}_{\text{pair}} = y_{\text{pair}}^*]$$

只奖励“正确且一致”的判断



总奖励（加权和）

$$R_{\text{total}} = \lambda_1 R_{\text{pair}} + \\ \lambda_2 R_{\text{point}} + \lambda_3 R_{\text{cons}}$$

●4.3 评估-LLM as a Judge角度

| Model | Agreement | Precision | Recall | F1 |
|------------------|-----------|-----------|--------|-------|
| LLaVA-1.5-7B | 42.07 | 39.57 | 30.29 | 25.73 |
| InternVL3-8B | 76.38 | 54.91 | 50.96 | 52.8 |
| GLM-4-9B | 68.24 | 46.68 | 45.54 | 46.05 |
| LLaVA-1.5-13B | 40.28 | 36.12 | 25.71 | 28.36 |
| InternVL3-14B | 75.23 | 58.26 | 50.03 | 53.7 |
| DeepSeek-V3-671B | 83.34 | 57.29 | 55.48 | 56.33 |
| FlexVL-7B* | 75.55 | 50.49 | 50.41 | 50.41 |
| JudgeLM-7B* | 46.57 | 41.43 | 30.89 | 35.27 |
| PandaLM-7B* | 53 | 54.53 | 53.77 | 51.3 |
| AgentJudge | 85.63 | 58.83 | 57.01 | 57.62 |

RewardBench

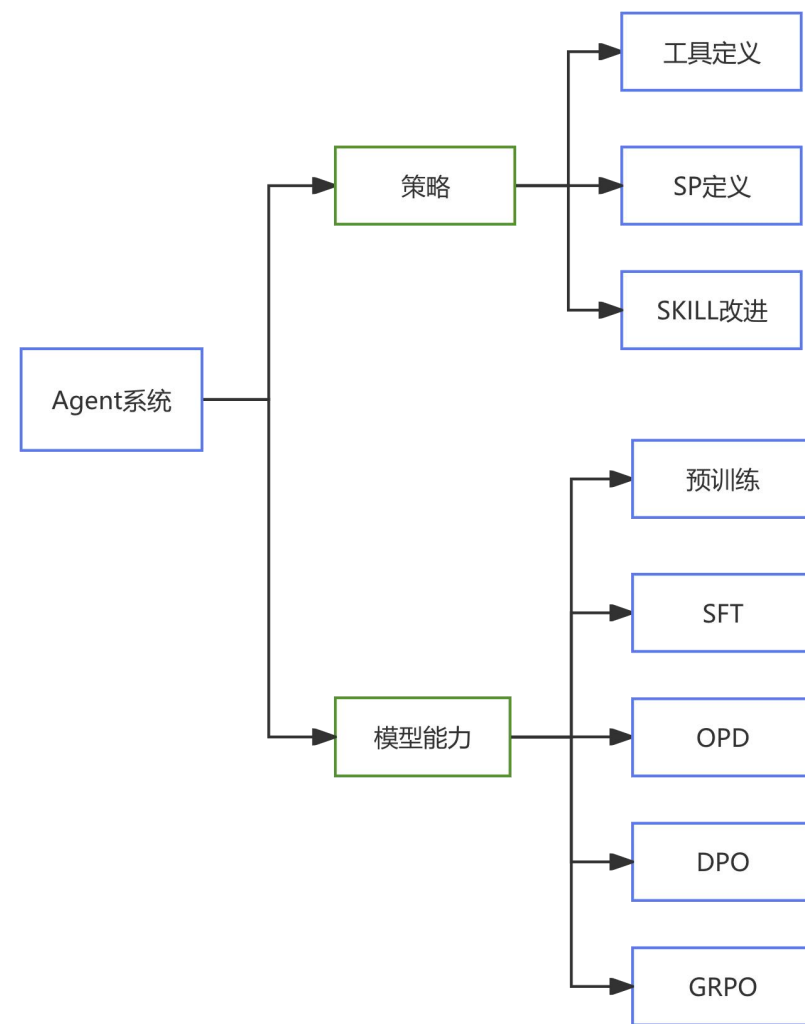
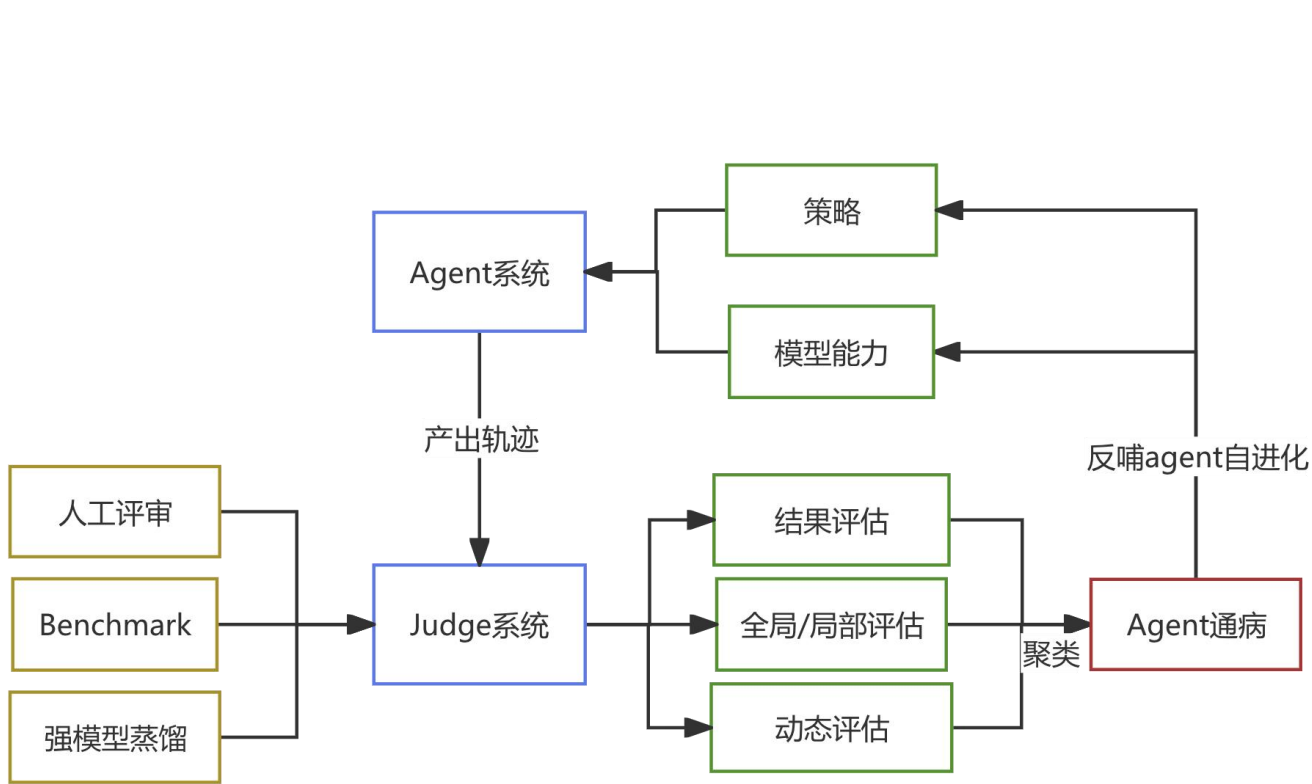
| Model | Agreement | Precision | Recall | F1 |
|------------------|-----------|-----------|--------|-------|
| GLM-4-9B | 70.02 | 69.86 | 57.69 | 59.55 |
| InternVL3-14B | 66.67 | 65.07 | 68.67 | 62.27 |
| LLaVA-1.5-7B | 37.13 | 40.11 | 39.24 | 28.98 |
| LLaVA-1.5-13B | 29.89 | 43.16 | 41.67 | 28.28 |
| Qwen2.5-72B | 72.8 | 66.25 | 63.6 | 64.58 |
| DeepSeek-V3-671B | 64.76 | 58.48 | 61.13 | 58.83 |
| FlexVL-7B* | 71.34 | 64.3 | 62.18 | 63 |
| JudgeLM-7B* | 66.77 | 63.83 | 71.95 | 63.92 |
| PandaLM-7B* | 65.82 | 44.56 | 48.93 | 46.24 |
| AgentJudge | 77.59 | 72.732 | 73.41 | 73.04 |

PandLM(Human)

●4.3 评估-Agent Judge角度

| 评测 | 数据集 |
|-----------------------------------|----------------------------------|
| General process verification | PRMBench, ProcessBench |
| Step-level agent assessment | AgentProcessBench, ToolPRMBench |
| Trajectory-level agent evaluation | AgentRewardBench, WebRewardBench |

●4.3 评估-反哺自进化（离线）



●4.3 评估-反哺自进化（在线）



●4.4 下一步计划

1. 分析实验Case，针对性补充数据、调优算法提高模型能力。
2. 完成剩余的实验。



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感谢各位老师的批评指正!